

A NUMERICAL STUDY OF CYCLIC MULTILINEAR MAPS, ORTHOGONAL PROJECTORS, AND MULTIREOLUTION TENSOR REPRESENTATIONS

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ABSTRACT. We report fourteen numerical experiments on a family of cyclically symmetric multilinear maps equipped with orthogonal projectors onto learned subspaces, together with a swept extension of four of them. Four results are exact identities that hold by construction and are reported as such: idempotence and self-adjointness of UU^* , cyclic invariance after exact symmetrisation, agreement of two independent implementations of a reduced tensor, and invariance of a corrected objective under a change of orthonormal basis. Four are negative results in the tested regime: a parameterised model of a commutator does not outperform the zero predictor in any of the fifteen trained checkpoints available; three independently trained resolutions exhibit no transport of learned subspaces, with principal angles close to $\pi/2$ throughout; the corresponding tensor factors do not persist across those resolutions; and three seeds reaching comparable training loss converge to subspaces that are pairwise nearly orthogonal. Two are methodological corrections found by the study's own controls: comparing orthonormal frames by unconstrained unitary Procrustes is vacuous, because the unitary group acts transitively on orthonormal frames of equal size; and an objective intended to be invariant under a change of basis was not, until it was summed over all columns. The remaining results are exploratory. A swept extension over 208 distinct configurations, each executed on a CPU and on a GPU, finds the GPU between 3.2 and 3.5 times slower than the CPU throughout $n \in \{12, 24, 48, 96\}$, and finds the observed associator ratio decreasing monotonically with ambient dimension, from 0.40 at $n = 12$ to 0.12 at $n = 96$, against a triangle bound of 2.

[2020]65F99, 65Y20, 15A69, 62-08 numerical linear algebra, principal angles, orthogonal projector, associator, cyclic symmetry, higher-order singular value decomposition, gauge invariance, identifiability, negative results

1. EXPERIMENTAL QUESTIONS AND STUDY DESIGN

1.1. The object under study. Fix an ambient dimension n , a reduced rank r and an arity a . The objects of the study are: a cyclically symmetric a -linear map μ on $\mathbb{C}^n$, stored in a rank- R canonical polyadic representation [9]; an isometry $U \in \mathbb{C}^{n \times r}$ with $U^*U = I_r$, and the associated orthogonal projector $P = UU^*$; and a reduced tensor obtained by contracting μ against U in every slot. Parameters are fitted by gradient descent against objectives built from a closure defect, an associator defect and a commutator residual, all defined in Section 2.

1.2. The fourteen questions. Each experiment answers one question. We list them here, and each subsequent section answers one. The parenthetical labels are the identifiers used in the accompanying software [3] and are given once, here, so that each result can be traced to its implementation; they are not used again.

- (Q1) Are the projector identities satisfied numerically? (implementation label: block A)
- (Q2) Can a parameterised model reproduce a commutator? (block B)
- (Q3) How do finite nested-commutator norms behave across families of projectors? (block C)

Date: 31 July 2026.

None of the results below has been reproduced by anyone other than the author, and no assessment of originality has been carried out. The finite-dimensional theory of projected composition trees [2] is a separate study and is not used here; see Remark 1.1.

- (Q4) Is thresholded spectral projection stable under perturbation? (block D)
- (Q5) Do learned subspaces transport across resolutions? (block E)
- (Q6) Is the fitted subspace locally identifiable, and invariant under a change of basis? (block F)
- (Q7) How large is the closure defect of a fitted map? (block G)
- (Q8) How large is the associator, against its triangle bound? (block H)
- (Q9) Do two independent implementations of the reduced tensor agree? (block I)
- (Q10) How should tensors at different resolutions be compared? (block J)
- (Q11) Is a fitted reduced tensor more compressible than a random one? (block K)
- (Q12) How much basis freedom remains after canonicalisation? (block L)
- (Q13) Do tensor factors persist across resolutions? (block M)
- (Q14) Does cyclic symmetrisation produce a cyclic map, and how large is a generalised Jacobi residual? (block N)

1.3. Reporting conventions. Three conventions are used throughout, and departures from them are the main defects we found in an earlier version of this work.

Identities by construction are labelled as such.: Several quantities vanish because of how they are defined, not because of anything that was fitted. Where this is the case we say so, and we do not report the vanishing as evidence about the fitted object.

A configuration is not an execution.: A *configuration* is one distinct mathematical setting: a choice of arity, dimension, rank, representation rank and seed. An *execution* is one run of one configuration on one device. Executions of the same configuration on different devices, and executions resumed from a checkpoint, are not independent observations and are never counted as such.

Statistical language is used only when a sampling model is stated.: Where no sampling population, statistic, uncertainty calculation and inferential procedure are specified, we write “observed over N sampled configurations” and not “statistically validated”.

Remark 1.1 (This study is separate from the projected-tree theory). The companion article [2] proves upper bounds for the error incurred by recursively projecting the intermediate outputs of a multilinear composition tree, and leaves the exact constants open. That study and this one share vocabulary — projectors, associators, multilinear maps — but not objects: no experiment below involves a composition tree or a recursively projected evaluation, and no numerical result below bears on the constants of [2]. In particular, the triangle bound of value 2 appearing in Section 10 is a bound on the norm of an associator of a single map, and is unrelated to the coefficient $k - 1$ of [2], which bounds the projected error of a signed combination of trees. The numeral is the same by coincidence.

2. DEFINITIONS

Let μ be an a -linear map on $\mathbb{C}^n$, $U \in \mathbb{C}^{n \times r}$ an isometry and $P = UU^*$.

Cyclic symmetrisation. Let σ rotate the input slots and $\Pi_{\text{cyc}} = \frac{1}{a} \sum_{j=0}^{a-1} \sigma^j$. Since σ acts unitarily on the coefficient tensor, Π_{cyc} is an orthogonal projector, and $\mu_{\text{cyc}} = \Pi_{\text{cyc}} \mu$ is cyclic. The *cyclic defect* of μ is $\|\mu - \Pi_{\text{cyc}} \mu\|_F$.

Closure defect. The normalised sampled closure defect is

$$\mathcal{E}_{\text{closure}} = \mathbb{E} \left[\frac{\|(I - P)\mu(PX_1, \dots, PX_a)\|^2}{\|\mu(PX_1, \dots, PX_a)\|^2 + \varepsilon} \right],$$

the X_i being independent standard complex Gaussian vectors. This is a sampled average, not the operator norm of the closure-leakage map, and it does not bound that operator norm.

Associator. With $A(x, y, z) = (x \circ y) \circ z - x \circ (y \circ z)$ for the binary product induced by anchoring one slot, the triangle inequality gives $\|A(x, y, z)\| \leq 2\widehat{M}^2 \|x\| \|y\| \|z\|$, where $\widehat{M}$ is a numerical estimate

of the operator norm of $\circ$. The *associator ratio* is the observed left-hand side divided by the right-hand side.

Commutator residual. For a Hermitian Δ , the identity $[\Delta, P] = K\Delta P - P\Delta K$ holds with $K = I$, and both sides have rank at most $2r$. The parameterised model is $C_\theta = U\Phi U^*\Delta K - K\Delta U\Phi U^*$ with Φ a free $r \times r$ matrix. The *unexplained relative residual* is $\|[\Delta, P] - C_\theta\|_F^2 / \|[\Delta, P]\|_F^2$; the zero predictor $C_\theta = 0$ gives the value 1.

Principal angles. Subspaces are compared by principal angles [1, 8]; $\pi/2 \approx 1.5708$ is complete orthogonality.

3. ORTHOGONAL PROJECTOR IDENTITIES

For any isometry U , the matrix $P = UU^*$ satisfies $P^2 = P$ and $P^* = P$ identically. Numerically, over five seeds and over all 416 executions of the swept extension of Section 17, the idempotence residual $\|P^2 - P\|_F / \|P\|_F$ stayed below 3.3×10^{-16} and the self-adjointness residual below 7.7×10^{-17} ; an exact $n = 2$ case agrees to a tolerance of 10^{-14} .

Remark 3.1. These identities hold by construction. They confirm that the implementation forms UU^* correctly and that the arithmetic is well conditioned. They are not evidence that the subspace $\text{ran } P$ is relevant to anything, and we never use them as such.

4. APPROXIMATION OF A COMMUTATOR BY A PARAMETERISED MODEL

4.1. The model is not capacity-limited in isolation. Training U against the unexplained residual alone drives that residual to approximately 10^{-9} across five seeds, matching a closed-form optimal- Φ ceiling. The model can therefore represent the target when nothing else competes.

4.2. It nevertheless fails in every trained checkpoint. In every one of the fifteen historical checkpoints carrying the relevant data, the model performed at or worse than the zero predictor.

Observation 4.1. *The proposed approximation did not outperform the zero approximation in any of the fifteen available trained models.*

This conclusion is restricted to the parameterisation, the objectives and the trained models examined here. It is not an impossibility statement for the model class.

4.3. Which factor is responsible. Table 1 reports seven training regimes, one run each, at $n = 16$, $r = 4$, representation rank 4, and 400 optimisation steps.

TABLE 1. Seven training regimes, one run per regime. “Unexplained” is the relative residual defined in Section 2; the zero predictor gives 1. Values are reproduced from the stored experiment record.

regime	unexplained	closure	associator
commutator objective alone	8.10×10^{-5}	0.310	464.1
with closure objective	3.51×10^{-4}	4.11×10^{-6}	74.6
with associator objective	8.49×10^{-3}	0.111	4.77×10^{-6}
all objectives jointly	6.00×10^{-3}	4.79×10^{-4}	1.07×10^{-6}
map frozen, projector trained	0.9925	1.56×10^{-2}	0.233
projector frozen, map trained	2.46×10^{-7}	0.738	580.9
staged: competing objectives, then commutator	4.65×10^{-4}	0.217	141.1

Two rows carry the interpretation. Freezing the map and training only the projector leaves the unexplained residual at 0.9925, barely below the zero predictor. Freezing the projector and training only the map — with U left at its random initial value — reaches 2.46×10^{-7} , which

is the numerical floor of that run and not exactly zero. Adding the associator objective to the commutator objective degrades the residual by about two orders of magnitude relative to training the commutator objective alone, whereas adding the closure objective does not.

Observation 4.2. *In these seven runs, the explanatory capacity resides in the parameters of the map rather than in the fitted subspace, and the commutator objective conflicts with the associator objective but not with the closure objective.*

This is a single run per regime. It is an exploratory result, not a controlled experiment with replicates, and it is offered as the best-supported reading of the seven runs rather than as an established mechanism.

Remark 4.3 (Relation to the underspecification literature). A pipeline is *underspecified* when it returns many distinct predictors with equivalently strong test performance which nonetheless behave differently in deployment [4]. The finding of Section 8 below — three seeds reaching comparable loss but converging to nearly orthogonal subspaces — is conceptually related to that phenomenon. The finding of the present section is different in mechanism: it concerns a conflict between two objectives within a single training run, rather than a multiplicity of equally good solutions. We record the connection as a conceptual one; the model, the objective and the conclusion analysed here are not those of [4], and this remark is not an assessment of originality.

5. FINITE NESTED-COMMUTATOR DIAGNOSTICS

We report Frobenius norms of finite nested commutators for four families of projectors: adversarially optimised, random, smooth, and spatially localised. The observed norms were 4.35, 3.84, 2.85 and 2.00 respectively.

That the adversarially optimised family produces the *largest* norms is the point of the experiment: it shows that the reported figures are not a favourable selection.

Remark 5.1 (Scope). This experiment computes finite nested-commutator norms and nothing else. It is not a test of membership in a pseudodifferential class and carries no microlocal-regularity content. Such a test would require symbol estimates of the kind recalled in the companion analytic article, and none is performed here.

6. STABILITY OF THRESHOLDED SPECTRAL PROJECTORS

Given a Hermitian matrix with a spectral gap, thresholding its spectrum yields a projector onto the corresponding invariant subspace. Perturbing by ϵH with H Hermitian, rank recovery and a Davis–Kahan-type bound [5] hold up to the point at which the gap closes. At $\epsilon = 1.2$ the gap closes from 1.0 to 0.51 and the rank is misrecovered.

Observation 6.1. *The tested perturbations confirm that the spectral-gap condition is necessary: rank recovery fails exactly when the gap closes.*

The experiment used genuine near-projectors, not only exact ones.

7. TRANSPORT OF LEARNED SUBSPACES ACROSS RESOLUTIONS

Three resolutions, $n = 12, 18$ and 24 , were independently initialised and independently trained. For each ordered pair, one frozen Gaussian-kernel lift operator maps the subspace of the lower resolution into the higher one, and the transported subspace is compared with the independently learned subspace there by principal angles. Two baselines are used: a random subspace of the same dimension, and a subspace obtained by interpolation.

All quantities reported in Table 2 lie between 1.331 and 1.553 radians, that is, between 76.3° and 89.0° . The transported angles themselves lie between 1.407 and 1.472. The trained lift improves

TABLE 2. Largest principal angle in radians, by resolution pair. Complete orthogonality is $\pi/2 \approx 1.571$. The trained lift beats both baselines in two of the three forward pairs.

pair	trained lift	random baseline	interpolation baseline	backward
12 $\rightarrow$ 18	1.410	1.480	1.527	1.331
12 $\rightarrow$ 24	1.472	1.416	1.468	1.540
18 $\rightarrow$ 24	1.407	1.553	1.446	1.370

on both baselines in two of the three forward pairs, and the margins where it does are 0.02 to 0.15 radians, small relative to how close every condition already is to complete orthogonality.

Observation 7.1. *The tested multiresolution models provided no evidence of transport of the learned subspace across the examined resolutions.*

The scope is three resolutions, one lift operator per pair, one training run per resolution. This is a negative result in the tested regime and not an impossibility statement.

8. LOCAL IDENTIFIABILITY AND INVARIANCE UNDER A CHANGE OF BASIS

8.1. An objective that was not invariant, and its correction. Any objective built from U should be invariant under $U \mapsto UV$ for unitary V , since UV spans the same subspace. The first version of the test objective used a single column of U and was therefore not invariant: the test detected a discrepancy of 0.87 % under such a change of basis. Summing over all r columns produces a Frobenius-norm quantity, which is exactly invariant under right multiplication by a unitary. This is an identity, and after the correction the discrepancy is zero to machine precision.

Remark 8.1. The defect was found by the experiment’s own required invariance control, not by inspection. Reporting it is the reason the control exists.

8.2. Curvature along the gauge direction. The exact Hessian and the generalised Gauss–Newton approximation $2J^\top J$ are computed separately and never conflated. The generalised Gauss–Newton matrix is positive semidefinite by construction; the exact Hessian is indefinite at a non-minimum, as expected. Along the infinitesimal direction generated by $U \mapsto UV$, the Hessian curvature is -3.5×10^{-18} , numerically flat, which is what invariance predicts.

8.3. Different seeds, nearly orthogonal subspaces. Three seeds all reach comparable near-zero training loss. The largest pairwise principal angle between the three converged subspaces is 1.55 radians, approximately 89° .

Observation 8.2. *Over the three seeds tested, comparable training loss was attained at subspaces that are pairwise nearly orthogonal. The subspace is not identified by the objective in the tested regime.*

Three seeds are three observations, not a statistical sample, and no interval is reported.

9. APPROXIMATE CLOSURE OF MULTILINEAR MAPS

The closure defect of a fitted map was evaluated over 2000 sampled configurations of the input vectors, and separately maximised by gradient ascent. The adversarial search found defects at least as large as the largest random sample in every case.

Observation 9.1. *The closure defect was observed over 2000 sampled inputs, with an adversarial maximum at least as large as the largest sampled value.*

Remark 9.2 (No sampling model is claimed). We do not describe this as a statistically validated result. Doing so would require stating the sampling population, the statistic, the number of independent observations, the uncertainty calculation, the inferential procedure and the limits of generalisation, and this study specifies none of them. No interval-arithmetic or sum-of-squares bound on the closure defect was derived.

10. BOUNDS FOR ASSOCIATORS

The triangle bound $\|A(x, y, z)\| \leq 2\widehat{M}^2\|x\|\|y\|\|z\|$ was never violated over 500 trials at a single configuration (300 random inputs and 200 adversarially optimised). The largest observed ratio was 0.452. The mean cosine between the two terms of the associator was 0.067: the terms are close to uncorrelated, rather than anti-aligned as the worst case of the triangle inequality would require.

Observation 10.1. *Across the configurations tested, the observed associator ratio remained well below the triangle bound, with a maximum of 0.957 over the swept extension of Section 17. No analytic improvement of the constant 2 is derived here, so the exact constant is undetermined.*

Section 17 extends this measurement over a grid and finds a pattern not visible at a single configuration.

11. EXTRACTION OF REDUCED TENSORS

Two independent implementations of the reduced tensor — one by explicit loops, one by a contraction exploiting the low-rank representation, including the cyclic average — agree to better than 10^{-10} in double precision and 10^{-4} in single precision. An exact rational case at $n = 2$, $r = 1$ matches a hand computation exactly, with no floating-point arithmetic involved.

This experiment tests extraction correctness only. It says nothing about whether the extracted tensor is meaningful.

12. GAUGE-INVARIANT COMPARISON OF TENSORS

Comparing two orthonormal frames $U_1, U_2 \in \mathbb{C}^{n \times r}$ by solving the unconstrained unitary Procrustes problem [10] over all of $U(n)$ is vacuous: the unitary group acts transitively on orthonormal r -frames in $\mathbb{C}^n$, so a unitary carrying U_1 to U_2 always exists, whatever the two frames are.

Observation 12.1. *An early version of the comparison tool used exactly this procedure and consequently reported independently drawn random tensors as equivalent. The defect was found by the experiment’s own required negative control, and was corrected by comparing subspaces with principal angles [1] instead.*

The corrected methodology reports four quantities separately — raw, Procrustes-aligned within the admissible group, permutation-aligned, and amplitude ratio — rather than collapsing them into one score. This is an improvement over comparing a single eigenbasis heuristic, and it is the methodology used in Sections 7 and 15.

13. MULTILINEAR SINGULAR-VALUE COMPRESSION

Using the multilinear singular value decomposition [6], a fitted reduced tensor requires mode ranks $[3, 4, 4]$ out of a full rank of 4 to capture 99% of its energy, against $[4, 4, 4]$ for a random tensor matched in norm.

Observation 13.1. *The fitted tensor is more compressible than the norm-matched random null in one of three modes, and equally compressible in the other two.*

This is a modest difference at a single configuration, and a single random tensor is not a null distribution.

14. CANONICAL REPRESENTATIVES AND RESIDUAL BASIS FREEDOM

After canonicalisation, the residual basis freedom is reported explicitly by the size of each spectral cluster. A deliberately constructed degenerate case, with eigenvalues $2, 1, 1$, is detected. Applying an arbitrary — not infinitesimal — unitary rotation within the two-dimensional degenerate eigenspace produces a frame that differs pointwise (Frobenius difference above 0.1) while spanning the same subspace (largest principal angle below 10^{-8}).

This is the concrete content of the statement that canonicalisation determines a representative only modulo the residual freedom.

15. PERSISTENCE OF TENSOR FACTORS ACROSS RESOLUTIONS

Applying the corrected comparison of Section 12 to the three independently trained resolutions of Section 7: the mode ranks required are inconsistent across the three resolutions ($\{3, 3, 4\}$, $\{3, 4, 4\}$, $\{3, 3, 3\}$), and the mean pairwise principal angle between corresponding modes is 1.01 radians.

Observation 15.1. *The tested multiresolution models provided no evidence of persistent tensor factors across the examined resolutions.*

One mode of the nine compared shows a near-exact match, 1.5×10^{-8} radians. We have no explanation for it. It is recorded here as an unresolved anomaly rather than absorbed into the summary above.

16. CYCLIC SYMMETRISATION AND A GENERALISED JACOBI RESIDUAL

The cyclic defect before symmetrisation is 4.60; after applying Π_{cyc} it is 8.2×10^{-33} , thirty-one orders of magnitude smaller.

Remark 16.1 (This is an identity). $\Pi_{\text{cyc}}\mu$ is cyclic by construction, because Π_{cyc} is the orthogonal projector onto the cyclically invariant subspace. The post-symmetrisation value is at the numerical floor because the identity is exact, not because a symmetry was learned. The figure is reported to confirm the implementation, and for no other purpose.

One exact, versioned generalised Jacobi expression — the full antisymmetrisation of the associator over the six permutations of three arguments — is implemented twice independently; the two implementations agree to 2.8×10^{-16} , and a deliberate sign mutation is detected. The supremum of the corresponding ratio is not shown to be bounded: the adversarial maximum is 5.98 at a single configuration against a mean of 0.43, and 4.65 to 5.92 across the 48 configurations of the pilot sweep. This ratio is undetermined.

The relation of this expression to Filippov’s fundamental identity [7] and to other conventions for generalised Jacobi identities has not been checked.

17. SWEPT EXTENSION OVER A GRID

The experiments above use a single configuration or a few seeds. Four of them — Sections 3, 9, 10 and 16 — were extended over a grid, on the hardware described in [3].

17.1. Design and accounting. There were no failures in either stage. Summed over the per-execution records, the wall time was 942.8 seconds for the pilot and 6300.8 seconds for the broad stage; the scheduler reported 945.5 and 6316.5 seconds respectively, the difference being scheduling overhead outside the timed region.

Two further stages — adaptive intensification of boundary configurations, and reruns with held-out seeds in extended precision with deterministic algorithms — were specified but not executed. Sections 4, 5, 6, 7, 8, 11, 12, 13, 14 and 15 were not extended over the grid and remain single-configuration evidence.

TABLE 3. Design of the swept extension. Each configuration was executed once on the CPU and once on the GPU. The two executions of a configuration are not independent observations.

stage	configurations	executions	grid
pilot	48	96	$a \in \{3, 4\}$, $n \in \{12, 24\}$, $r \in \{3, 6\}$, $R \in \{4, 8\}$, 3 seeds
broad	160	320	as above with $n \in \{12, 24, 48, 96\}$, 5 seeds
total	208	416	

TABLE 4. Mean wall time per execution, in seconds, over the 160 configurations of the broad stage. Each mean is over 40 executions.

n	CPU	GPU	ratio
12	9.44	30.64	3.24
24	9.31	30.80	3.31
48	9.83	34.50	3.51
96	7.90	25.09	3.17

17.2. Device throughput.

Observation 17.1. *Over the tested range $n \in \{12, 24, 48, 96\}$, the GPU was between 3.17 and 3.51 times slower than the CPU for these four experiments, with no trend towards crossover.*

The ratio is stable across an eightfold range in ambient dimension, so the effect is not a small- n artefact. The practical consequence for further work on these four experiments, on this hardware, is to use the CPU. No asymptotic-complexity statement and no comparison between hardware platforms follows from this measurement.

17.3. The associator ratio decreases with ambient dimension. Over the 96 executions of the pilot the observed associator ratio ranged from 0.130 to 0.957 with mean 0.334; over the 320 executions of the broad stage it ranged from 0.042 to 0.957 with mean 0.240. Broken down by ambient dimension, the mean ratio was 0.404 at $n = 12$, 0.257 at $n = 24$, 0.182 at $n = 48$ and 0.118 at $n = 96$.

Observation 17.2. *The mean observed associator ratio decreases monotonically with ambient dimension over the tested range, moving further from the triangle bound as n grows.*

The single-configuration value 0.452 of Section 10, measured at $n = 16$, is consistent with this trend rather than an outlier.

17.4. Reproducibility of the generalised Jacobi ratio. The adversarial maximum of the generalised Jacobi ratio reproduced between 4.65 and 5.92 over the 48 pilot configurations, against 5.98 at the original single configuration. The finding is stable across configurations.

18. PROVENANCE OF THE HISTORICAL EVIDENCE

An earlier body of evidence for these questions was re-examined before this study was carried out, and the re-examination changed what that evidence supports.

Observation 18.1. *The nineteen distinct historical runs are all exploratory: every one records an evaluation mode of **screening**, including the run directories whose names suggest otherwise. The nine runs recorded in the run log share a single seed and form one continuous chain of non-strict resumptions rooted at a single bootstrap; they are not nine independent trials. Two distinct*

script hashes appear across the chain, so the script changed during the campaign without this being recorded.

The scoring procedure used at the time computed a global score as the mean of per-experiment scores, with an inconclusive experiment contributing half credit and an experiment that never ran contributing full credit; no failing state existed. A mean over such a vocabulary cannot decrease when an experiment fails to run, so we do not use it. Throughout this article each experiment is reported separately and no global score is formed.

19. LIMITATIONS

- (1) The negative results of Sections 4, 7, 8 and 15 are established for the objectives actually tested. For tractability these used a closure-only or two-term objective rather than the full weighted objective of the historical campaign, and Section 4 itself shows that the choice of objective changes the outcome materially.
- (2) The swept extension covers four of the fourteen experiments. The remaining ten are single-configuration evidence and were not run on the GPU at all.
- (3) Two further sweep stages were specified and not executed.
- (4) Single random tensors are used as nulls in Section 13. A single draw is not a null distribution.
- (5) Timing measurements come from one machine and one software stack.
- (6) No result here has been reproduced by anyone other than the author.
- (7) No assessment of originality has been carried out. A bounded literature search located directly relevant prior work for three of this study’s own methodological choices — a protocol of the same conservative design in an unrelated domain, established provenance tooling for run-lineage reconstruction, and the underspecification literature [4] for the phenomenon of Section 8. None of the methodological corrections described above should be read as claimed novel.

20. SUMMARY

TABLE 5. Summary of the fourteen experiments and the swept extension.

Question	Conclusion	Evidence
Projector identities (§3)	satisfied	identity by construction
Commutator model (§4)	not supported in the tested models	negative result
Nested commutators (§5)	adversarial family gives the largest norms	exploratory
Spectral projector stability (§6)	gap condition is necessary	counterexample
Subspace transport (§7)	not observed	negative result
Identifiability (§8)	subspace not identified	negative result
Basis invariance (§8)	holds after correction	identity by construction
Closure defect (§9)	measured	observed over 2000 samples
Associator bound (§10)	constant undetermined	exploratory
Reduced tensor (§11)	implementations agree	exact check
Tensor comparison (§12)	unconstrained Procrustes is vacuous	counterexample
Compressibility (§13)	modest, in one mode	exploratory
Residual basis freedom (§14)	quantified	identity by construction
Factor persistence (§15)	not observed; one anomaly	negative result
Cyclic symmetrisation (§16)	identity	identity by construction
Generalised Jacobi ratio (§16)	undetermined	exploratory
Device throughput (§17)	CPU faster throughout	negative result
Associator trend (§17)	decreases with n	exploratory

Four of the questions received negative answers in the tested regime, two produced methodological corrections that were found by the study's own controls, four are identities that hold by construction, and the remainder are exploratory. No global score is formed from these, and no result is described as established beyond the regime in which it was measured. Submission of this study is deferred pending independent reproduction and an assessment of originality, neither of which has been carried out.

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